Mr SHARE ONLINE REVISION ft. TUKS. t; 1~2 testing

General Certificate of Education Advanced Level

STATISTICS **6046/1**

PAPER 1

SOLUTION 2023 (ZIMSEC EXAM) 3 hours

Additional materials:
- Answer paper
- Graph paper
- List of formulae MF7
- Electronic calculator (Non-programmable)

Time 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name in the spaces provided on the answer sheet/answer booklet.

Answer ***all*** questions.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given to 2 significant figures.

INFORMATION TO CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part of question.

The total number of marks for this paper is 120.

The use of a scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 7 printed pages and 1 blank page.

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1.

(a)

(i) **Direct observation**

- By observing the rain as it enters directly into the rain gauge, collecting data and recording the data manually/ electronically using electronic rain gauges.

(ii) **Advantage**

- The respondent is unaware of being observed and therefore behaves in a natural way.

Disadvantage

- It is a passive form of data collection as there is no opportunity to investigate the behavior further.

2.

(a)

(i) **Mutually exclusive events**

- Events which cannot both occur/take place at the same time/ together.
- If A and B are mutually exclusive events then;

  $P(A \cap B) = 0$

  $P(A \cup B) = P(A) + P(B)$

(ii) **Independent events**

- Events which have no effect on each other/ the occurrence of one event has no effect on the outcome of the other.
- If **A** and **B** are independent events then;

  $P(A \cap B) = P(A) \times P(B)$

(b)

(i) Prime numbers $= 2, 3, 5, 7$

$$P(prime\ number) = \mathbf{\frac{4}{9}}$$

Odd numbers $= 1, 3, 5, 7, 9$

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(ii)

$$P(O|P) = \frac{(O \cap P)}{P(P)}$$

$$= \frac{\frac{3}{9}}{\frac{4}{9}}$$

$$= \mathbf{\frac{3}{4}}$$

3. $n = 15$ $\sum x = 724.25$

$\sum x^2 = 40852.5375$

Mean;

$$(\bar{x}) = \frac{\sum x}{n} = \frac{724.25}{15}$$

$$= 48.283 \ldots$$

$$= \$\mathbf{48.28}$$

Standard deviation;

$$(\delta) = \sqrt{\frac{40852.5375}{15} - \left(\frac{2897}{60}\right)^2}$$

$$= 19.804 \ldots$$

$$= \$\mathbf{19.80}$$

4. Total $(n) = 52$

Spades $(S) = 13$

Not spades $(S') = 52 - 13 = 39$

$$\therefore P(S = 4) = \frac{{}^{13}_{4}C \times {}^{39}_{2}C}{{}^{52}_{6}C}$$

$$= \frac{529815}{20358520}$$

$$= \mathbf{0.026} \ \ (\mathbf{2}\ \boldsymbol{s.f})$$

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OR;

Spades	Hearts	Diamonds	Clubs
4	2	-	-
4	-	2	-
4	-	-	2
4	1	1	0
4	1	0	1
4	0	1	1

$$P = \left(\frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} \times \frac{10}{49} \times \frac{13}{48} \times \frac{12}{47} \times \frac{6!}{2! \times 4!} \times 3\right)$$
$$+ \left(\frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} \times \frac{10}{49} \times \frac{13}{48} \times \frac{13}{47} \times \frac{6!}{4!} \times 3\right)$$
$$= \mathbf{0.026} \ \ (\boldsymbol{2\ s.f})$$

5.

(a) **Permutations** $({}^{n}_{r}\boldsymbol{P})$ – is the number of ways of arranging a subset of r objects from a group of n objects where the order of selection is important.

Combination $({}^{n}_{r}\boldsymbol{C})$ – the number of ways of selecting a subset of r objects from a group of n objects where the order of selection is not important.

(b) Number of books $= 9$

From same author $= 4\ (B)$

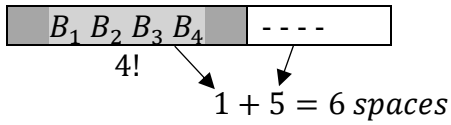
Not from the same author $= 5(B')$

(i) Number of ways $= 9! = 362880$

OR;

Number of ways $= {}^{9}_{9}P = 362880$

(ii)



Number of ways $= 4! \times 6! = 17280$

NB: Treat 4 books as one, and it can be arranged in 4! ways, now we have 6 books which can be arranged in 6!. Hence in total we have 6!4! ways.

OR;

Number of ways $= {}^4_4P \times {}^6_6P = 17280$

6.

(a) Bayes Theorem

- A mathematical formula for determining conditional probabilities of events which describes the probabilities of occurrence of an event related to any condition
- Describes the probability of an event, based on prior knowledge of conditions that might be related to the event.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

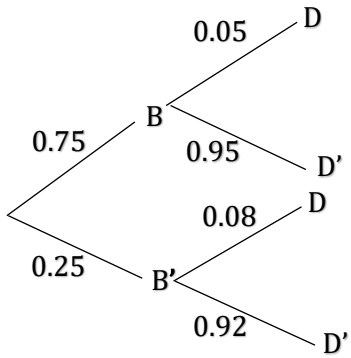
A, B – events

$P(A|B)$ - probability of A given B is true

$P(B|A)$ – probability of B given A is true

$P(A), P(B)$ – independent probabilities of A and B

(b)



$$P(B|D) = \frac{P(B \cap D)}{P(D)}$$

$$= \frac{0.75 \times 0.05}{(0.75 \times 0.05) + (0.25 \times 0.08)}$$

$$= \frac{0.0375}{0.0575}$$

$$= \mathbf{\frac{15}{23}} \ \boldsymbol{or} \ (\mathbf{0.65})$$

7. Let X be the r.v 'drivers for passenger vehicle who pay a fine'

$$X \sim Bin(15; 0.24) \quad q = 0.76$$

(i) $P(X = 3) = {}^{15}_{3}C \times 0.24^3 \times 0.76^{12}$

$= 0.2335 \ldots$

$= \mathbf{0.23} \ \ (\mathbf{2} \ \boldsymbol{s.f})$

(ii) $P(X \geq 3) = 1 - P(X \leq 2)$

$= 1 - [0.76^{15} + ({}^{15}_{1}C \times 0.24 \times 0.76^{14}) \times ({}^{15}_{2}C \times 0.24^2 \times 0.76^{13})]$

$= 0.735 \ldots$

$= \mathbf{0.74} \ \ (\mathbf{2} \ \boldsymbol{s.f})$

8.

(a) Let 'H' be the r.v 'the number of heads obtained'

$$P(H = 0) = TTT = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Amount awarded $= 2(0) = 0$

Win or loss $= 0 - 4 = -4$

$$P(H = 1) = HTT = \left(\frac{1}{2}\right)^3 \times \frac{3!}{2!} = \frac{3}{8}$$

Amount awarded $= 2(1) = 2$

Win or loss $= 2 - 4 = -2$

$$P(H = 2) = HHT \times \frac{3!}{2!} = \left(\frac{1}{2}\right)^3 \times 3 = \frac{3}{8}$$

Amount awarded $= 2(2) = 4$

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Win or loss $= 4 - 4 = 0$

$$P(H = 3) = HHH = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Amount awarded $= 2(3) = 6$

Win or loss $= 6 - 4 = 2$

h	0	1	2	3
$P(H = h)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$
Win /loss	-4	-2	0	2

Expected win/ loss

$$= \frac{1}{8}(-4 + (-6) + 2)$$

Or

$$= \frac{1}{8}(-4) + (-2)\left(\frac{3}{8}\right) + 2\left(\frac{1}{8}\right)$$

$$= -1$$

$$= \$\mathbf{1} \ \ (\boldsymbol{loss})$$

(b) No, I would not advise him because the game is unfair since there is only a $\frac{1}{8} \times 100\% = 12.5\%$ chance of winning/ he will lose \$1

9.

(a) $n = 12$ $\quad\quad \sum y = 790$

$\sum x = 838$

$$(\bar{x}) = \frac{\sum x}{n} = \frac{838}{12}$$

$$= 69.833 \ldots$$

$$= \mathbf{70} \ \ (\mathbf{2}\ \boldsymbol{s.f})$$

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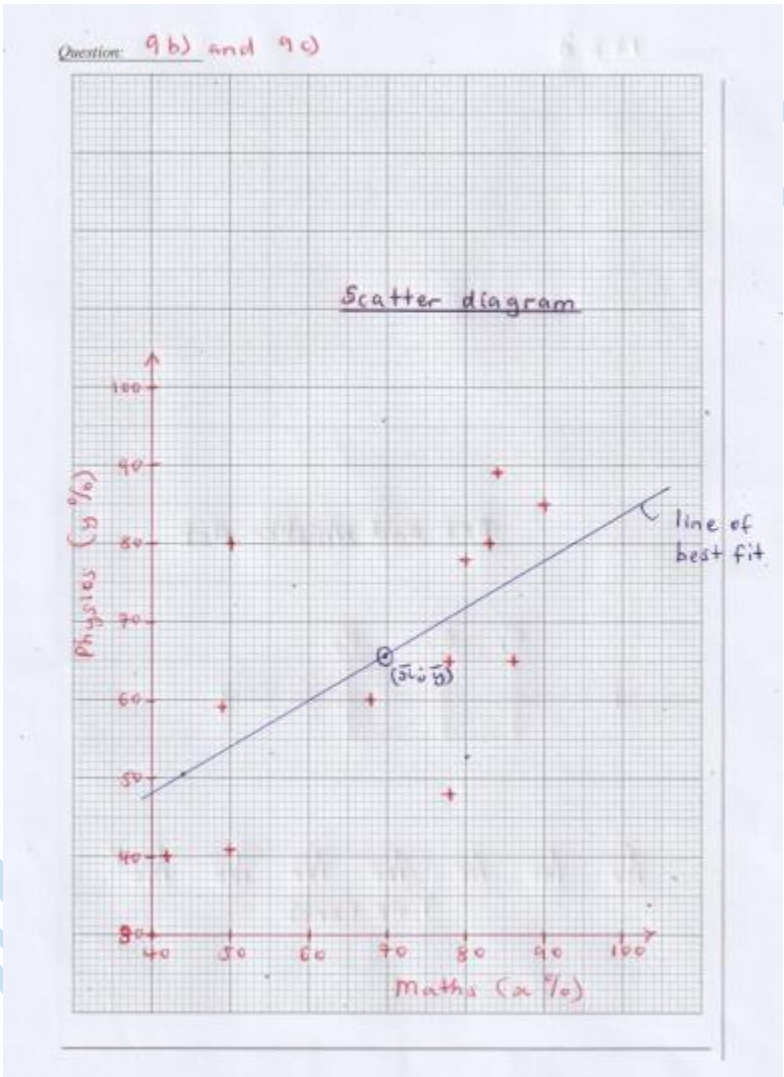
$$(\bar{y}) = \frac{\sum y}{n} = \frac{790}{12}$$
$$= 65.833 \ldots$$
$$= \mathbf{66} \ \ (\mathbf{2}\ \boldsymbol{s.f})$$

(b) Drawing a scatter diagram on the graph;

(c) Drawing of line of best fit on graph;



NB; Using $(\bar{x}; \bar{y}) = (69.8; 65.8)$ and any other point for example $(44; 50.5)$

10.

(a) If n is large and $np > 5$ and $nq > 5$

(b) Let X be the r.v 'students who enjoy basketball'

$$X \sim Bin(50; 0.2) \quad q = 0.8$$

$\mu = np = 50 \times 0.2 = 10$

$\delta^2 = npq = 50 \times 0.2 \times 0.8 = 8$

$$\therefore X \sim N(10; 8)$$

$P(X > 15) \quad \rightarrow \quad P(X > 15.5)$

$$= P\left(Z > \frac{15.5 - 10}{\sqrt{8}}\right)$$

$$= P(Z > 1.945)$$

$$= 1 - \emptyset(1.945)$$

$$= 1 - 0.9741$$

$$= \mathbf{0.0259}$$

NOTES;

- ✓ Normal distribution can be used to approximate binomial distribution when n is large and $np > 5$, and $nq > 5$.
- ✓ Remember to apply continuity correction when approximating binomial/ poison distribution **(discrete)** by a normal distribution **(continuous).**

Discrete	Continuous
$P(X > a)$	$P(X > a + 0.5)$
$P(X \geq a)$	$P(X > a - 0.5)$
$P(X < a)$	$P(X < a - 0.5)$
$P(X \leq a)$	$P(X < a + 0.5)$
$P(X = a)$	$P(a - 0.5 < X < a + 0.5)$

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11.

(a) $k + 2k + 3k + 4k = 1$

$$10k = 1$$

$$\boldsymbol{k = \frac{1}{10}}$$

(b) $E(X) = \frac{1}{10}(1 + 4 + 15 + 24)$

$$= \mathbf{4.4} \ \ \boldsymbol{or} \ \ \left(\mathbf{4\frac{2}{5}} \ \ \boldsymbol{or} \ \ \mathbf{\frac{22}{5}}\right)$$

(c) $Var(X) = E(X^2) - [E(X)]^2$

$$= \frac{1}{10}(1 + 8 + 75 + 144) - (4.4)^2$$

$$= 22.8 - 19.36$$

$$= \mathbf{3.44} \ \ \boldsymbol{or} \ \ \mathbf{3\frac{11}{25}} \ \ \boldsymbol{or} \ \ \mathbf{\frac{86}{25}}$$

12.

(a) **Time series;**

A set of numeric data/ sequence of observations of a random variable taken over time at equally spaced points /regular intervals (hourly/ weekly/ monthly/ quarterly/ yearly) and arranged in a chronological time order of their occurrence.

(b) **3 types of variations**

- Cyclical variations
- Seasonal variations
- Irregular/ random variations

(c) Number of 4 – point MA = 3 [i.e (6 – 4)+ 1 = 3]

13.

(a) Let X be the r.v 'the number of clients with houses built with the strong cement'

H_0: $p = 0.95$

H_0: $p \neq 0.95$

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If H_0 is true; $X \sim Bin(20; 0.95)$

We perform a two tailed test at 5% level of significance.

We reject H_0 if $P(X \leq x) < 2.5\%$ or $P(X \geq x) < 2.5\%$

Test value; $x = 16$ $(lower\ tail)$

Computation; $P(X \leq 16)$

$= 1 - P(X \geq 17)$

$= 1 - [{}^{20}_{17}C \times 0.95^{17} \times 0.05^3 + {}^{20}_{18}C \times 0.95^{18} \times 0.05^2 + {}^{20}_{19}C \times 0.95^{19} \times 0.05 + 0.95^{20}]$

$= 0.016 < 0.025$

We reject H_0 and conclude that at 5% level of significance there is insufficient evidence to support the manufacturer's claim.

(b) No, because there is no evidence that the cement has a greater success rate in providing strong walls.

14.

(a) $X \sim Bern(p)$ $\quad q = 1 - p$

Standard deviation;

$$\delta = 0.3$$

$$\sqrt{pq} = 0.3$$

$$p(1-p) = 0.3^2$$

$$p - p^2 = 0.09$$

$$p^2 - p + 0.09 = 0$$

$$p = \frac{1 \pm \sqrt{0.64}}{2}$$

$$p = \frac{1 \pm 0.8}{2}$$

$$p = 0.9 \ \ or \ \ 0.1$$

$$\therefore \boldsymbol{E(X) = 0.9 \ \ or \ \ 0.1}$$

NOTES;

If $X \sim Bern(p) \quad q = 1 - p$

- $E(X) = p$ and $Var(X) = pq \rightarrow \text{s.d}, \delta = \sqrt{pq}$

Difference between Binomial and Bernoulli distribution;

- In binomial distribution the number of trials n is finite and $n > 1$.
- In Bernoulli distribution the number of trial n is fixed to 1 only.

✓ Tossing a fair coin once (Bernoulli distribution) but tossing a fair coin (2, 3 or more times) is a binomial distribution.

(b) $X \sim Geo(0.7) \quad q = 0.3$

(i) Mode $= 1$

NB: The mode of a geometric distribution is always 1.

(ii) $E(X) = \frac{1}{p}$

$= \frac{1}{0.7}$

$= \mathbf{1\frac{3}{7}} \ \ \boldsymbol{or} \ \left(\frac{\mathbf{10}}{\mathbf{7}} \ \ \boldsymbol{or} \ \ \mathbf{1.4}\right)$

(iii) Standard deviation;

$$\delta = \sqrt{\frac{1-p}{p^2}}$$

$$= \sqrt{\frac{0.3}{0.7^2}}$$

$$= 0.782 \ldots$$

$$= \mathbf{0.78} \ \ (\mathbf{2}\ \boldsymbol{s.f})$$

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15.

(a) Conditions for Poisson distribution;

- Events must occur singly
- Events must occur uniformly
- Events must occur at a constant rate
- Events must occur independently
- Events must occur randomly

(b) If $X \sim Po(4.5)$

(i) Mean;

$$E(X) = \beta$$
$$= \mathbf{4.5}$$

Standard deviation;

$$\text{S.d}, \delta = \sqrt{Var(X)} = \sqrt{\beta}$$
$$= \sqrt{4.5}$$
$$= 2.121 \ldots$$
$$= \mathbf{2.1} \ \ (\mathbf{2}\ \boldsymbol{s.f})$$

(ii) $P(X \geq 6) = 1 - P(X < 6)$

$$= 1 - e^{-4.5}\left(1 + 4.5 + \frac{4.5^2}{2!} + \frac{4.5^3}{3!} + \frac{4.5^4}{4!} + \frac{4.5^5}{5!}\right)$$

Or;

$$= 1 - e^{-4.5}\left(1 + 4.5 + \cdots + \frac{4.5^2}{5!}\right)$$
$$= 0.297 \ldots$$
$$= \mathbf{0.30} \ \ (\mathbf{2}\ \boldsymbol{s.f})$$

(iii) By trial and error;

$$P(X < 6) = P(X \leq 5) = 0.7029$$
$$P(X < 5) = P(X \leq 4) = 0.5321$$

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$$\therefore a = 5$$

NB: (strictly by trial and error)

16.

Mid-point (t)	5	15	30	60
Frequency (f)	2	f	11	4

$$\sum ft = 580 + 15f$$
$$\sum f = 17 + f$$

(a) $\bar{t} = \frac{\sum ft}{\sum f}$

$$27.5 = \frac{580 + 15f}{17 + f}$$
$$27.5(17 + f) = 580 + 15f$$
$$467.5 + 27.5f = 580 + 15f$$
$$12.5f = 112.5$$
$$\boldsymbol{f = 9}$$

(b) Median $= O_{me} + \frac{c\left[\frac{n}{2} - F(<)\right]}{f_{me}}$

- O_{me} – lower limit of median class
- n – total frequency
- $F(<)$ – cumulative frequency of class before median class
- c – class width of median class
- f_{me}- frequency in median class

$$= 20 + \frac{20\left[\frac{26}{2} - 11\right]}{11}$$
$$= 20 + \frac{40}{11}$$
$$= 23.636 \ldots$$
$$= \boldsymbol{24\ mins\ (2s.f)}$$

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Comment;

50% of the students spent less than nearly 24 minutes and the other 50% spent more than nearly 24 minutes.

(c) Standard deviation;

$$\delta = \sqrt{\frac{\sum ft^2}{\sum ft} - (\bar{t})^2}$$

$$= \sqrt{\frac{26375}{26} - 27.5^2}$$

$$= 16.067 \ldots$$

$$= \mathbf{16\ mins\ \ (2s.f)}$$

17.

(a)

(i)

Stem	Leaf
13	2 9
14	0 5 8 8
15	1 4 7
16	2 2 6
17	1 8
18	2

Key; 16|2 means 162 sec

(ii) Advantages of stem and leaf

- Shows all the data values
- No data is lost
- Shows the shape of the distribution

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(b)

(i) $Q_1 = \frac{16}{2} = 4^{th}\ value$

$= \mathbf{145}$

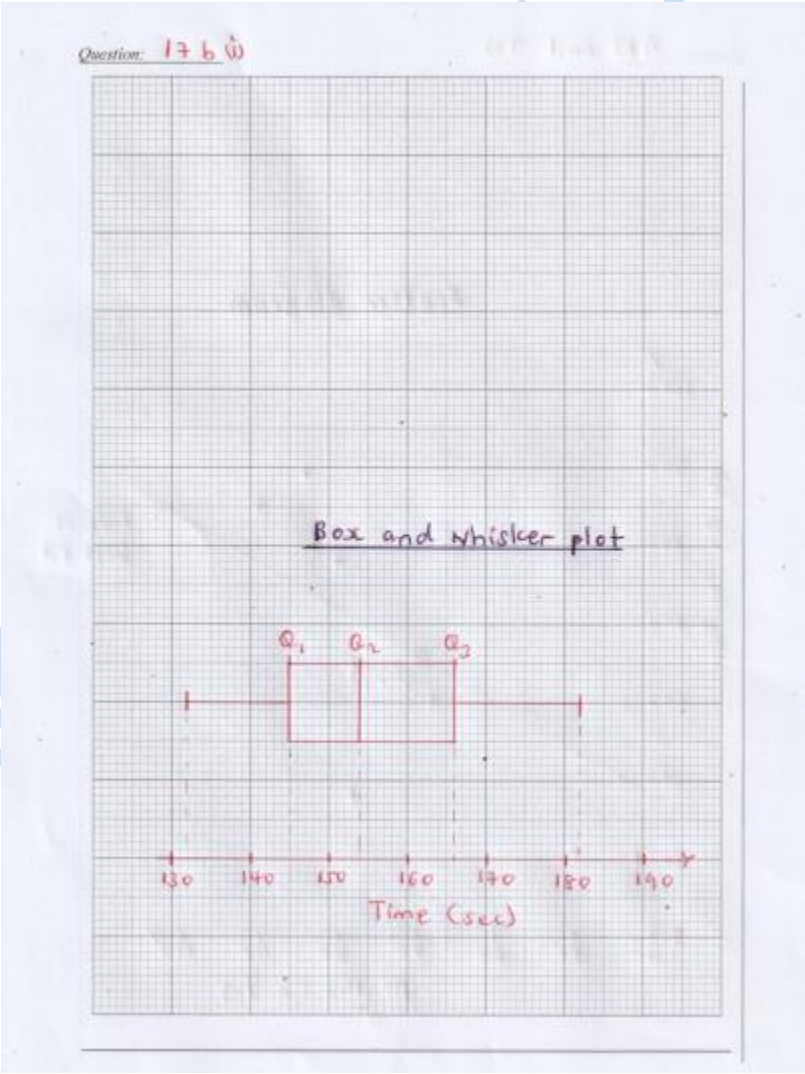
$Q_2 = \frac{16}{2} = 8^{th}\ value$

$= \mathbf{154}$

$Q_3 = \frac{3(16)}{4} = 12^{th}$

$= \mathbf{166}$

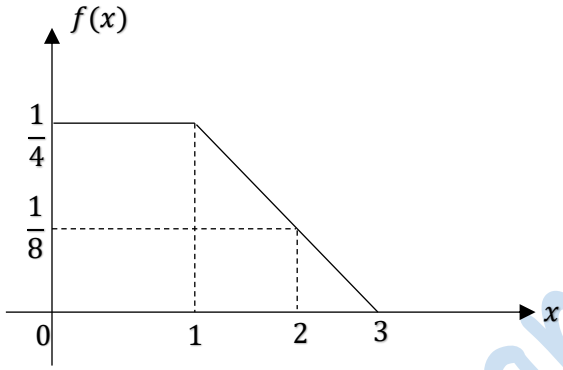
NB; drawing of the and whisker plot on the graph



(ii) Disadvantages of box and whisker plot;

- Cannot be drawn where exact data is not known
- Cannot be drawn on rounded data
- It only shows minimum and maximum values and quartiles

18.



(a) Taking;

$$\int_0^1 \frac{1}{4}\,dx + \int_1^3 k(3-x)\,dx = 1$$

$$\frac{1}{4}[x]_0^1 + k\left[3x - \frac{x^2}{2}\right]_1^3 = 1$$

$$\frac{1}{4} + k\left[\left(9 - \frac{9}{2}\right) - \left(3 - \frac{1}{2}\right)\right] = 1$$

$$\frac{1}{4} + k(2) = 1$$

$$2k = \frac{3}{4}$$

$$\boldsymbol{k = \frac{3}{8}}$$

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Now;

Taking;

$$f(x) = k(3-x)$$

$$\frac{1}{4} = k(3-1)$$

$$\frac{1}{4} = 2k$$

$$\boldsymbol{k = \frac{1}{8}}$$

NOTE;

IF WE USE THE VALUE OF $k = \frac{3}{8}$, THE FUNCTION $f(x)$ WILL NOT BE CONTINUIOUS AT $x = 1$ i.e $\left[\frac{3}{8}(3-1) = \frac{3}{4},\ \text{instead of } \frac{1}{4}\right]$.

BUT IF WE USE THE VALUE OF $k = \frac{1}{8}$, THE FUNCTION $f(x)$ WILL BE CONTINUOUS AT $x = 1$ i.e $\left[\frac{1}{8}(3-1) = \frac{1}{4}\right]$, WHICH IS INDEED TRUE FOR $y = \frac{1}{4},\ 0 < x < 1$. HENCE THE VALUE OF $k = \frac{1}{8}$

Also refer to ZIMSEC June 2013 P4 #5, and J Crawshaw Exercise 6d # 4 and 9

(b) Mean;

$$E(X) = \int_0^1 \frac{1}{4}x\ dx + \int_1^3 \frac{1}{8}x(3-x)dx$$

$$= \frac{1}{4}\left[\frac{x^2}{2}\right]_0^1 + \frac{1}{8}\left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_1^3$$

$$= \frac{1}{4}\left[\frac{1}{2}\right] + \frac{1}{8}\left[4.5 - \frac{7}{6}\right]$$

$$= \boldsymbol{\frac{13}{24}}\ \ \boldsymbol{or}\ \ (\mathbf{0.54})$$

(c)

(i)

$$P(0.8 < X < 2) = \int_{0.8}^{1} \frac{1}{4} dx + \int_{1}^{2} \frac{1}{8}(3 - x)\, dx$$

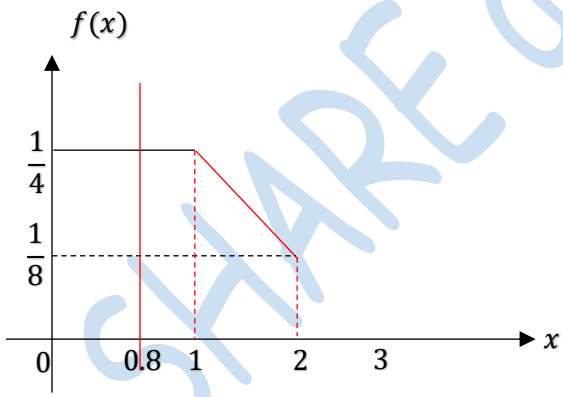
$$= \left[\frac{1}{4}x\right]_{0.8}^{1} + \frac{1}{8}\left[3x - \frac{x^2}{2}\right]_{1}^{2}$$

$$= \left[\frac{1}{4} - \frac{1}{5}\right] + \frac{1}{8}\left[4 - \frac{5}{2}\right]$$

$$= \mathbf{\frac{19}{80}} \ \ \boldsymbol{or} \ \ (\mathbf{0.2375})$$

OR;

Area of rectangle + trapezium



$$P(0.8 < X < 2) = \left(\tfrac{1}{4} \times 0.2\right) + \tfrac{1}{2}\left(\tfrac{1}{4} + \tfrac{1}{8}\right) \times 1$$

$$= \mathbf{\tfrac{19}{80}} \ \ \boldsymbol{or} \ \ (\mathbf{0.2375})$$

(ii) $P[X > E(X)] = P\left(X > \frac{13}{24}\right)$

$$= \left[\frac{1}{4}\left(1 - \frac{13}{24}\right) + \left(\frac{1}{2} \times 2 \times \frac{1}{4}\right)\right]$$

$$= \mathbf{\frac{35}{96}} \ \ \boldsymbol{or} \ \ (\mathbf{0.36})$$

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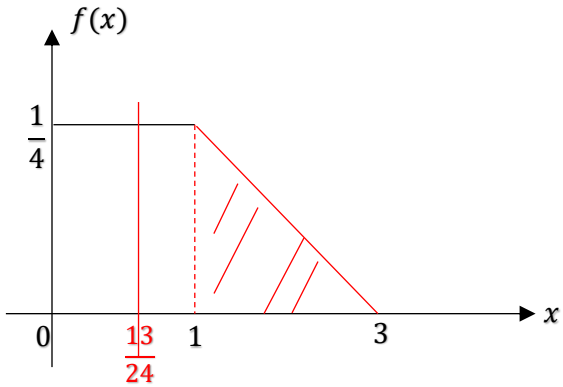
NB; Area of rectangle + triangle, $\frac{1}{2}bh$



OR;

$$P[X > E(X)] = \int_{\frac{13}{24}}^{1} \frac{1}{4}dx + \int_{1}^{3} \frac{1}{8}(3-x)dx$$

$$= \frac{1}{4}[x]_{13/24}^{1} + \frac{1}{8}\left[3x - \frac{x^2}{2}\right]_1^3$$

$$= \left[\frac{1}{4} - \frac{13}{96}\right] + \frac{1}{8}\left[\left(9 - \frac{9}{2}\right) - \left(3 - \frac{1}{2}\right)\right]$$

$$= \frac{11}{96} + \frac{1}{8}\left[\frac{9}{2} - \frac{5}{2}\right]$$

$$= \mathbf{\frac{35}{96}} \ \boldsymbol{or} \ [\mathbf{0.36} \ \ (\mathbf{2}\boldsymbol{s.f})]$$

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FEEL FREE TO CONTACT ME FOR ANY ADJUSTMENTS, CLARIFICATIONS AND ASSISTANCE!

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"Concept before anything!", Author

Proverbs 11 vs. 2

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Tutors in:

- O – A Level Accounting
- O – A Level Mathematics
- O - A Level Additional/ Further Mathematics
- Business Mathematics
- Engineering Mathematics
- Actuarial Mathematics
- Quantitative Techniques and Methods

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NB: If you click any of the links below and it fails to direct you straight to the respective group, then first copy the whole link of any group and, place it in any WhatsApp/Telegram chat/group, then click again and join!

WhatsApp group 1 link: https://chat.whatsapp.com/GkRjwDjJr808QbpRRZ3wkf

WhatsApp group 2 link: https://chat.whatsapp.com/Hv3KgLep5etBUdE1RjPkDF

WhatsApp group 3 link: https://chat.whatsapp.com/BTOJO6v6yi5HGWH2LfCg7t

WhatsApp group 4 link: https://chat.whatsapp.com/IzJ7csROuVhANBmHkCifAT

WhatsApp group 5 link: https://chat.whatsapp.com/DJeLdwJWCmi6J8PeNPoLOk

Telegram group link: t.me/shareALmathematics

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